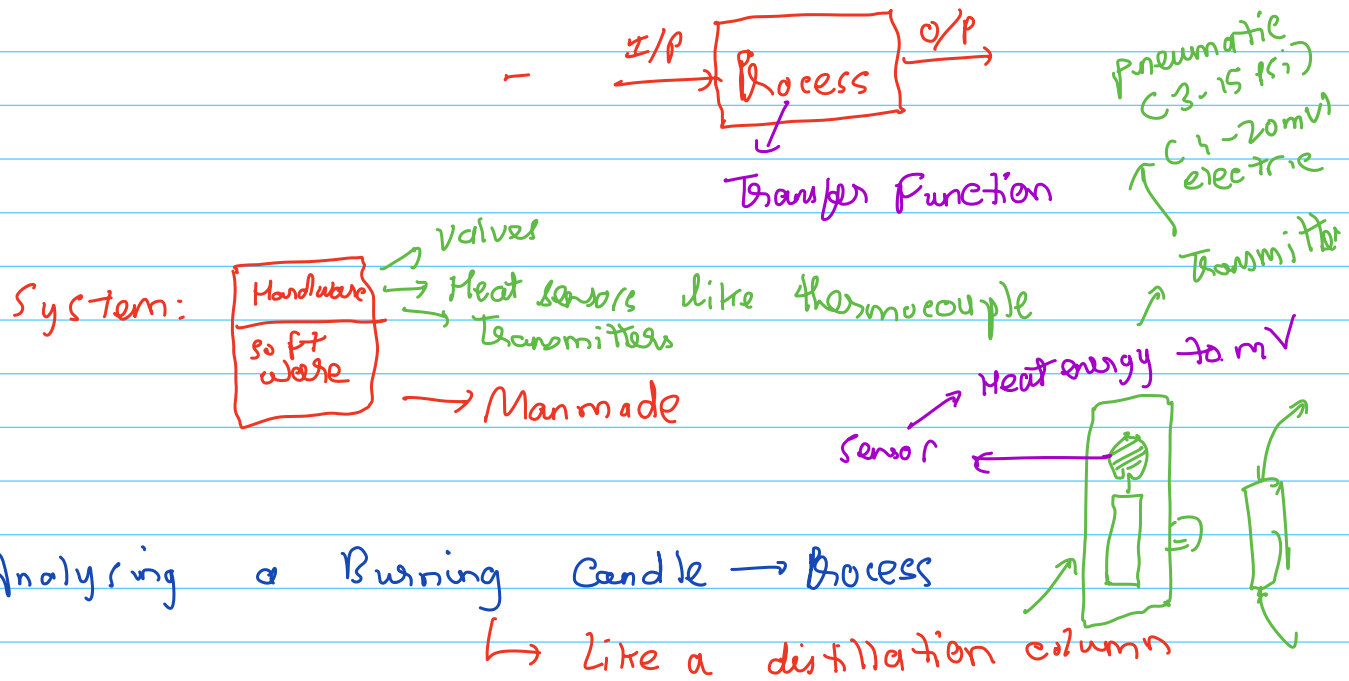


8/1/24

RTIPS



- System is required to monitor, control or optimize a process. Transmitter converts it to psi, and then it goes to a valve.



No, "intelligence" required for H/W and Data Acquisition

- Intelligence: If the software is intelligent, you can optimise the system. Software constitutes the "intelligent" part of the system.

Process remains as it is. You can't change the system

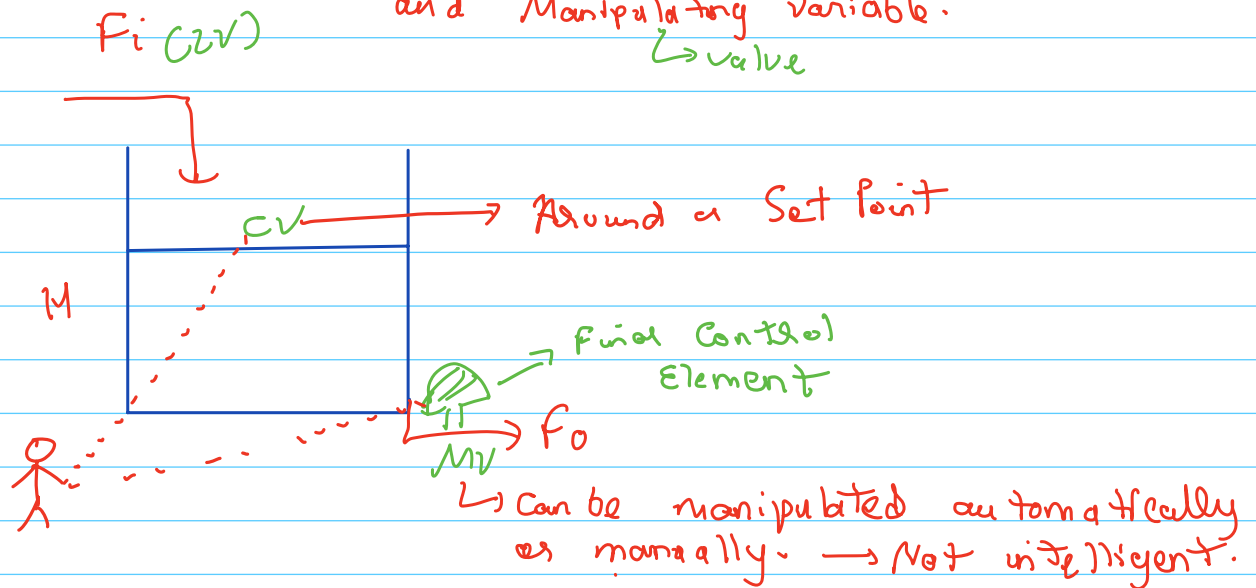
- Real Time:-

Offline Data: Real, but not live. Used for training / backtesting. → Fast

Online :- One way communication. You are on live data but cannot interfere. → Till the data is

Real time:- You are live and can work on real data. You can interfere with the data.

Has load variable, control variable and Manipulating variable.
 $\rightarrow$ value



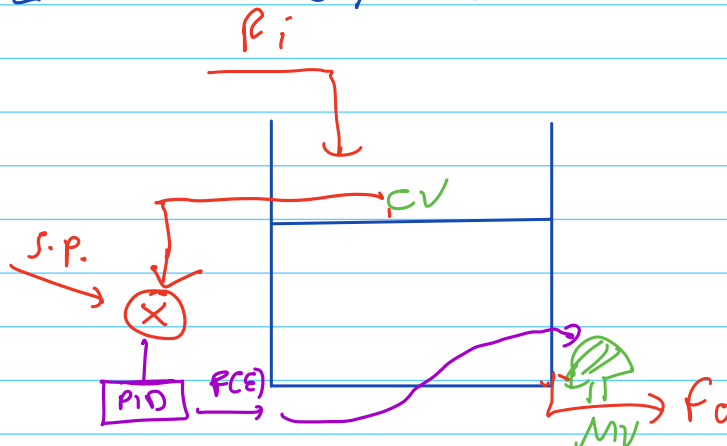
Intelligence is imparted through either mathematical or AI based modelling.

9/1/24

Purdue University

AUTOMATION HIERARCHY

- Level-0 :- Sensor / PCE $\rightarrow$ They must be provided with the equipment itself.
- Level-1 :- LCL / Scada $\rightarrow$ Minimises human intervention



Scada $\rightarrow$ TAG
 $\rightarrow$ Data Structure

H. PV

H. SP

H. HI

H. LO

Planned variable

Even though this is a closed variable, S.P. has to be manually assigned. Hence it still requires manual intervention. S.P. need not always be constant too.

and downloads to

- Level 2:- It calculates the Level 1 S.P. and sets it for Level 1. It is the 'Intelligence' level. Purely software

Mathematical Modelling

AI Based

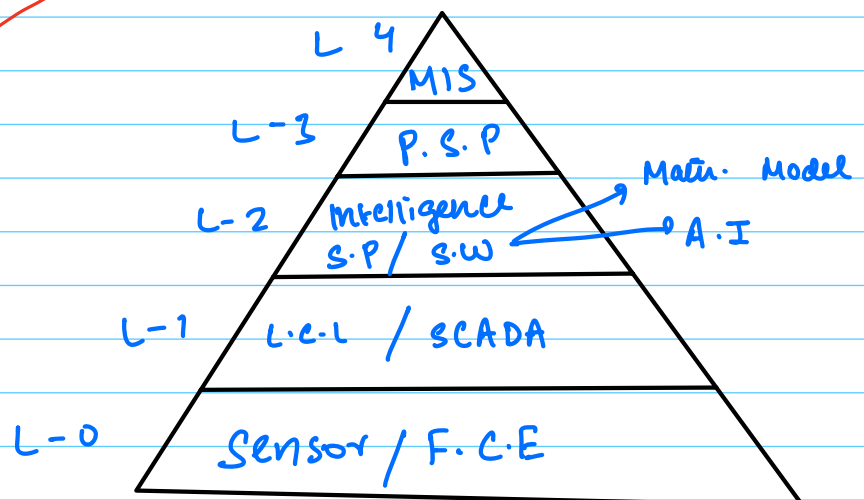
- There is always a lag in the data provided to the neural networks. The time lag must be adjusted. You need domain expertise (e.g. Blast Furnace) by normalising and shifting the time scale by some factor.

Time	P _i	P ₀	M
-	⊖	-	-
-	-	-	-
-	-	⊖	-

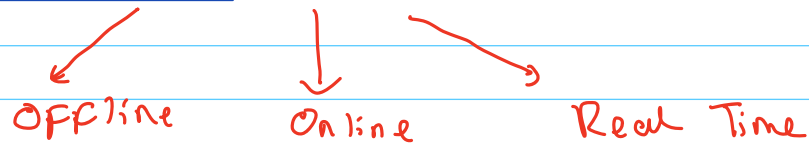
- Level 3: production scheduling & planning (PSP)

Management

- Level 4: MIS



* MATHEMATICAL MODELLING



- Equations, assumptions, B.C., I.C., algorithms, simulations

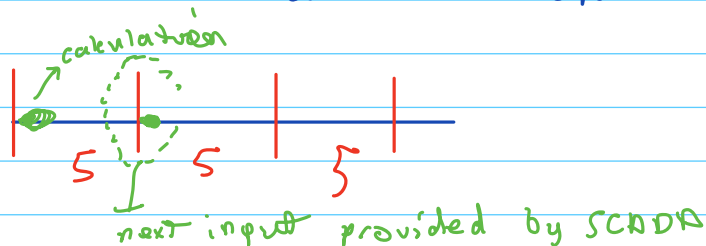
- Equations

- Algebraic
- ODE
- PDE

$$F_i - F_o = \frac{dH}{dt} (CPJ)$$

Offline: Back testing
/ R.T.

Online: Scada provides data to the models from the process, you calculate the output and then measure it with the actual output.



Model should predict M_{next} after 5 seconds, and it can then be used as the S.P. for level 1.

$$F_i - F_o = \frac{dH}{dt} (CPJ)$$

$$\frac{dH}{dt} \equiv \frac{H^{n+1} - H^n}{\Delta t}$$

S.P.

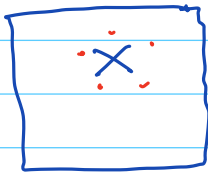
Now back-calculate to control F_o .

This model is a CONTROL MODEL.

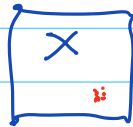
A SIMULATION MODEL will actually calculate H .



SENSORS



ACCURACY

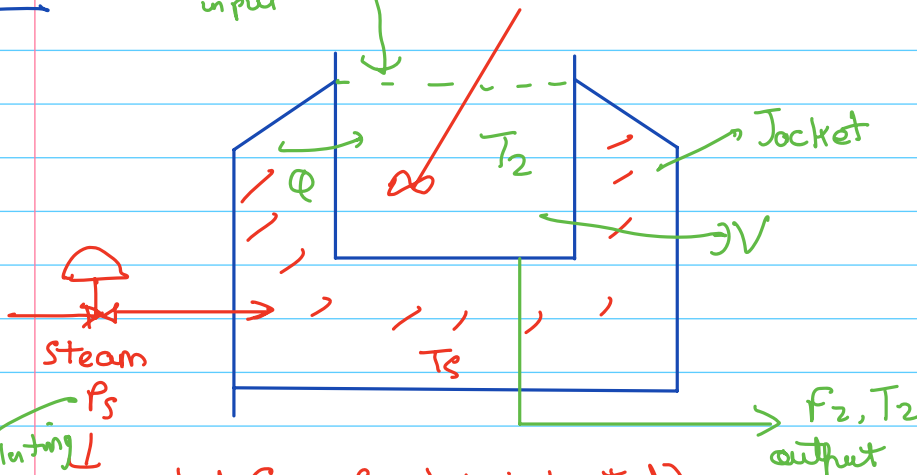


PRECISION

- Good Sensors should be accurate and precise.

15/1/24

P_1, T_1
input



manipulating
variable

Saturated (at P_s , it is saturated)

output

- ① variables
- ② Assumptions
- ③ Eqs
↓
mass, energy, momentum

Heat Transfer v/s

Thermodynamics

↓
No work

↓
work

↓
Saturated steam
is used
↓
as latent heat > sensible
heat

Heat can never be 100%
converted to work

↓
Superheated steam
is used.

↓
You want condensation
to happen to
utilise latent heat.

eg> Turbines
steam (superheated)
↓
Work
to prevent
condensation
of steam

Assumptions:-

- No heat loss
- Perfectly mixed
- Physical properties are constant

Equations

- Mass Balance:

$$F_1 - F_2 = \frac{dV}{dt}$$

$$\textcircled{1} \equiv F_1^{n+1} - F_2^{n+1} = \frac{V^{n+1} - V^n}{\Delta t}$$

we compare V^{n+1} to actual V^{n+1}
 $\nearrow$ Real Time
 $\nwarrow$ Real Time

Implicit

Explicit
 $V^{n+1} = \dots$

- Heat Balance:

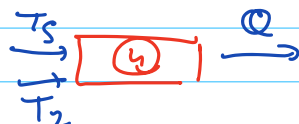
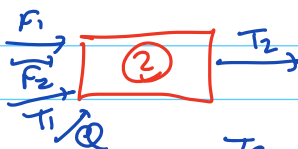
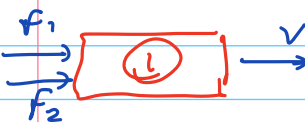
$$F_1 C_p T_1 - F_2 C_p T_2 + \frac{Q}{J} = \frac{d}{dt} (V C_p T_2) \quad \textcircled{2}$$

Service Heat

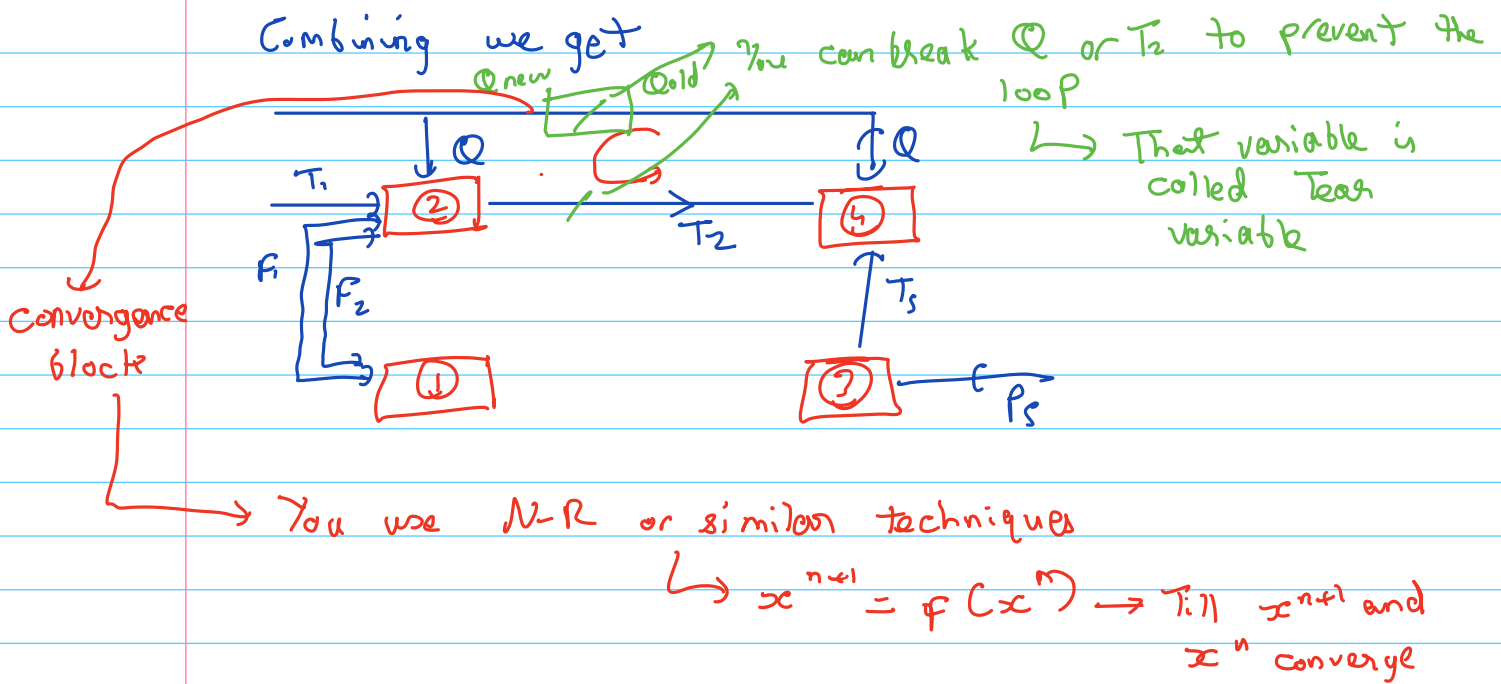
- Jacket: Steam Table
 $\hookrightarrow T_s = f(P_s)$ ③

- Jacket: Heat transfer
 $Q = UA(T_s - T_2)$ ④

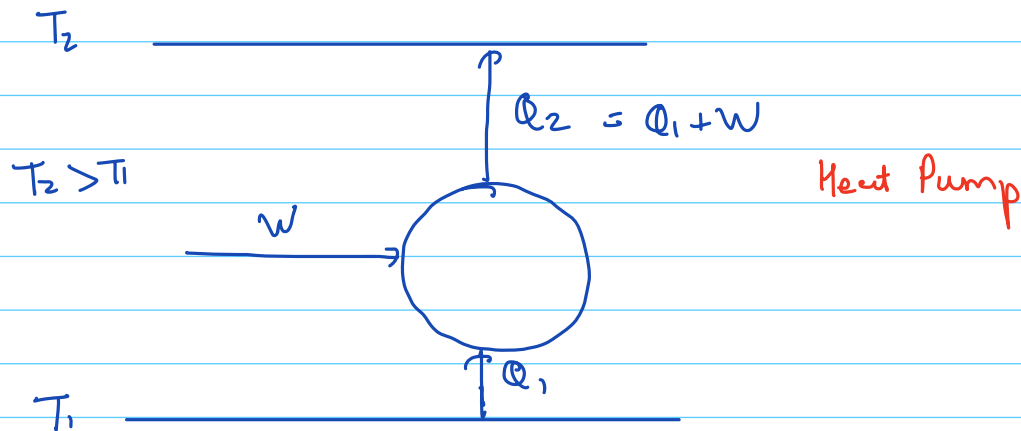
* INFORMATION Flow DIAGRAM

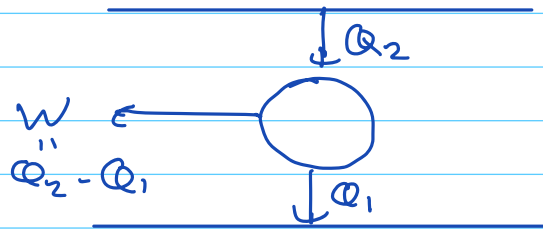


Not selecting V as it is an output for ①



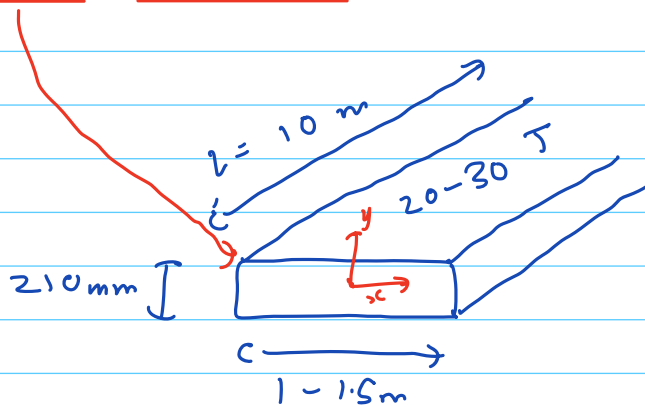
16/1/24



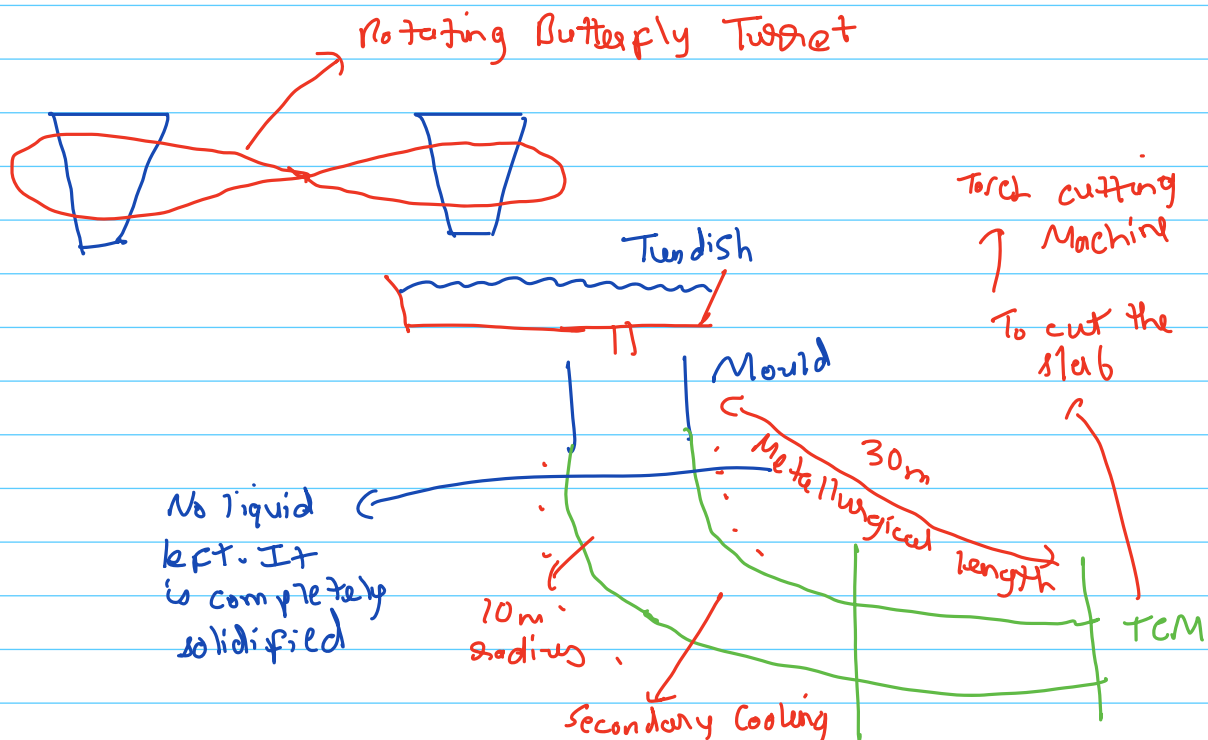


Heat Engine

* SLAB CASTER - CONTINUOUS



- Prepared by L-D process → Liquid steel from liquid iron
oldest steel plants in the world



Shell needs to be thick enough to support the peristaltic pressure.

- Speed at which casting is taking place $\rightarrow$ Casting Speed
1-1.5 m/min
Beyond V_{cast} , metallurgical length and breakout speed increases

$$\underbrace{k \left(\frac{\partial^2 T}{\partial x^2} \right) + k \left(\frac{\partial^2 T}{\partial y^2} \right)}_{\text{conductive}} = \rho C \frac{\partial T}{\partial t} + \underbrace{\rho V_{cast} \frac{\partial T}{\partial x}}_{\text{convective/advection}}$$

23/1/24

SLAB CASTER EQUATION

$$\underbrace{\rho C_p \frac{\partial T}{\partial t}}_{\text{transient}} + \underbrace{\rho V_{cast} \frac{\partial T}{\partial x}}_{\text{advection}} = k \underbrace{\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)}_{\text{conduction}}$$

$$\underbrace{V_{cast}}_{1.5 \text{ m/min}} \times \underbrace{Area}_{210 \text{ mm} \times 1.5 \text{ mm}} \times \underbrace{Density}_{7400 \text{ kg/m}^3} \rightarrow m^3 (\text{kg/s})$$

$$\Rightarrow \dot{m} = 58.275 \text{ kg/s} \approx 58.3 \text{ kg/s.}$$

Now

$$\lambda = 272 \text{ kJ/kg}$$

$$C_p = 750 \text{ J/kgK}$$

$$T_{start} = 1750^\circ \text{C (l.)}$$

$$T_{end} = 600^\circ \text{C (solid)}$$

$$\dot{m} (C_p \Delta T) = 58.3 (-272 + 0.75(550))$$

$$= 3906.35 \text{ kJ/s}$$

$$= 39.9 \text{ MW}$$

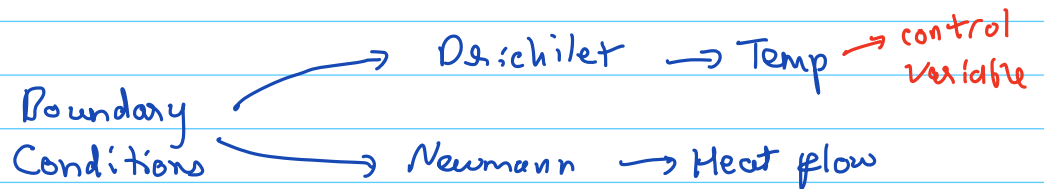
10% heat is extracted in mould

↓
50% heat extracted from secondary cooling zone

30% C??

- "As the residence time ↑, it will ↑ the cooling efficiency".
- As the dimension of the slab increases, we have to reduce V_{cast} , as it will ensure proper cooling of the slab.

BOUNDARY CONDITIONS



- Empirical Relation of water Spray:

$$q = A(\dot{w})^n$$

$\swarrow$ Provided by manufacturer
 $\searrow$ water flow rate
 $\nearrow$ lit/min/m² in research paper

- For the mould,
heat extracted by the mould.

$$q = A - B\sqrt{t}$$

↗ residence time

- Casting Speed (V_{cast}) is the load variable.
- Manipulated variable $\rightarrow$ is the water flow rate.
- Temp - control variable
- Objective : To maintain a particular temperature profile.
Any objective of level 2 is to calculate the SP and send it to level 1.
- Model calculates the ideal water flow rate which will provide the setpoint temp. and will send it to level 1. (ie PI; PID) to maintain the set point.
- Cooling Rate governs the mechanical property of steel.
- For a **GRADE** of Steel, the 'cooling rate is specified'
 $\rightarrow$ we know the temp. profile.
We apply this profile to the model and then we will calculate the flux and hence the w value is obtained and sent to level 1.
- Simulation Mode } Secondary cooling
Control Mode } zone only
- In the mould there is no control. The cooling water flow rate is already defined.
- Simulation Mode: Takes the ideal temp. profile as the input and predicts the flux.
ie $\rightarrow$ we get the value of w .

* SOLVING THE EQUATION

Alternate Direction Implicit (ADI)

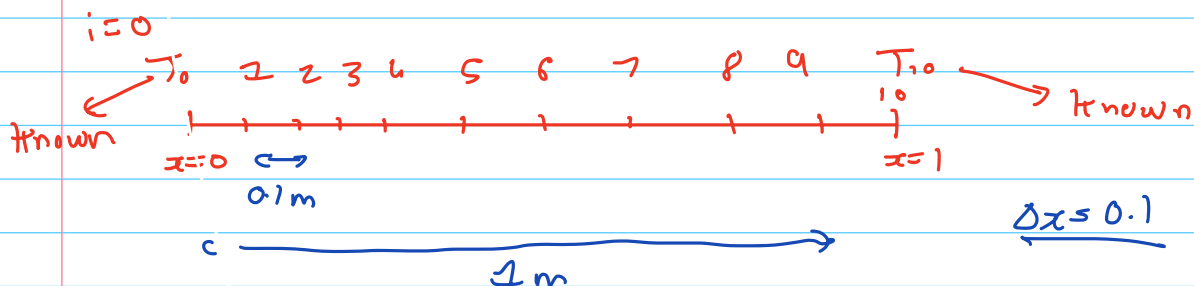
$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

$$\alpha = \frac{k}{\rho C_p}$$

FTCS

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \alpha \left(\frac{T_{i+1}^{n+1} - 2T_i^{n+1} + T_{i-1}^{n+1}}{(\Delta x)^2} \right)$$

- Explicit has inherent stability issue
- Implicit scheme is intrinsically stable.
- Instability issues can be handled using small time steps.
- Implicit Schemes involved a system of eqns



$i=1$

$$\frac{T_1^{n+1} - T_1^n}{\Delta t} = \alpha \left(\frac{T_2^{n+1} - 2T_1^{n+1} + T_0}{(\Delta x)^2} \right)$$

$$\frac{T_9^{n+1} - T_9^n}{\Delta t} = \alpha \left(\frac{T_{10}^{n+1} - 2T_9^{n+1} + T_8^{n+1}}{(\Delta x)^2} \right)$$

$$\begin{bmatrix} x & x & 0 & 0 & 0 & \dots \\ 0 & x & x & x & \dots \\ \vdots & \dots & \dots & 0 & 0 & x & x \end{bmatrix}_{9 \times 9} \begin{bmatrix} T_1 \\ T_2 \\ \vdots \\ T_9 \end{bmatrix} = \begin{bmatrix} \end{bmatrix}$$

$$A T = b \Rightarrow T = A^{-1} b$$

If we have

$$b T_{i-1}^{n+1} + d T_i^{n+1} + a T_{i+1}^{n+1} = c$$

$$a = b = -\frac{q \Delta t}{(\Delta x)^2}$$

$$d = 1 + 2 \frac{q \Delta t}{(\Delta x)^2}$$

$$d = 1 + 2\lambda$$

Thomas Algorithm:

$$\begin{bmatrix} d & a & \dots \\ b & d & a & \dots \\ \vdots & \dots & \dots & d \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_N \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{bmatrix}$$

$$d_j' = d_j - \frac{b_j}{d_{j-1}} a_{j-1}$$

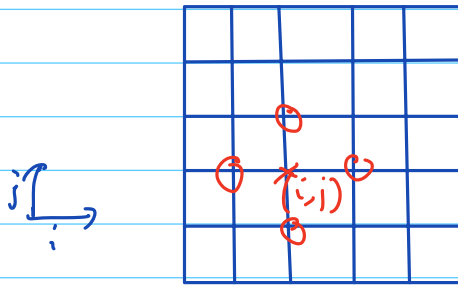
$$j = 2, 3, \dots, N$$

$$c_j = c_j - \frac{b_j}{d_{j-1}} c_{j-1}$$

Then, back substituting $u_N = c_N / d_N$

$$\Rightarrow u_k = \frac{c_k - a_k u_{k+1}}{d_k}$$

$$k = N-1, N-2, \dots, 1$$



5 unknowns for every eqn
for a 2-D system.

- For 2-D and 3-D we need to reduce the dimensions.
This can be done using alternate deduction implicit.
- Taking implicit in one direction at a time
 - using line by line TDMA
- Horizontal Sweep (B-T)

Vertical Sweep (L-R)

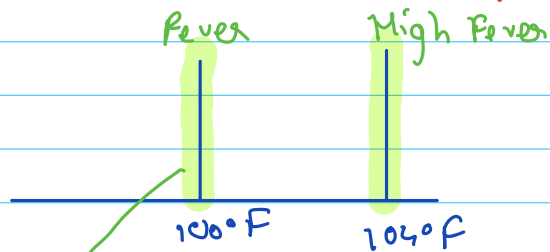
M_1	(B-T)	} 1 cycle → in one time step
V_1	(R-L)	
M_2	(T-B)	
V_2	(L-R)	

5/2/24

*

FUZZY LOGIC
↙ ↓
opposite: crisp cause → effect

Crisp: Has hard and sharp lines.

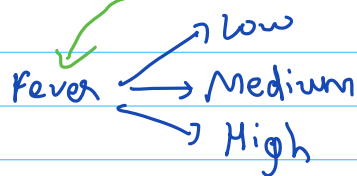


Fuzziness, as 99.99°F or 100.01°F can also be considered as fever.

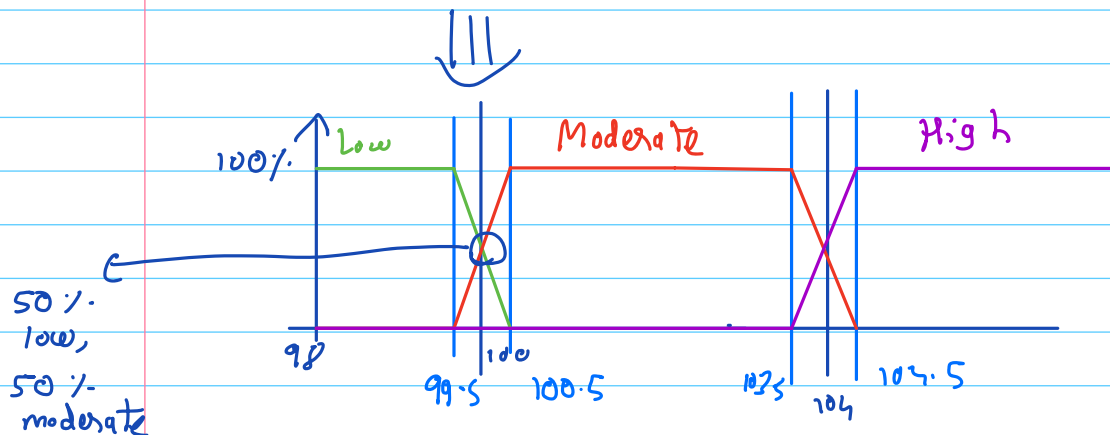
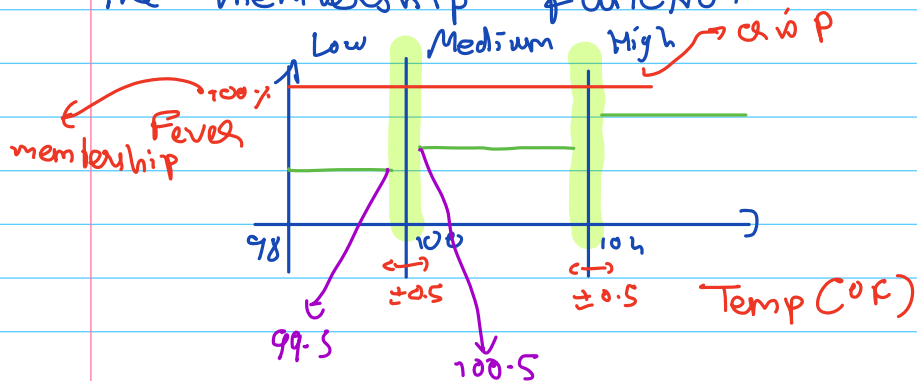
- Our brain can process imprecise, vague, incomplete logic.
fuzzy processing
- Crisp logic can process only precise information, like numbers. → Like robots.
- $A \rightarrow \{1, 2, 5, 10\}$ → Crisp Set. No ambiguity
↳ 100% certain
- Fuzzy A → $\{ \underline{1}, \underline{2}, \underline{5}, \underline{10} \}$ → Here, we have partial membership
0.2 0.4 0.8 0.1
↳ 20% member ↳ 40% member..
- Membership Function :-

Low fever, High fever :- Linguistic variables

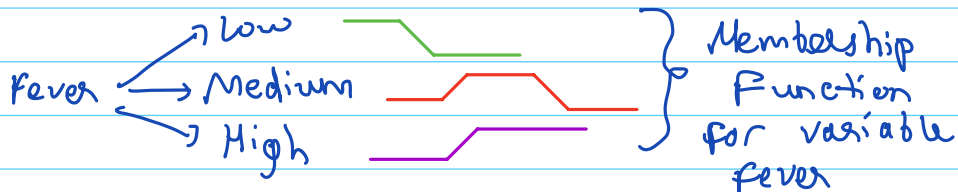
Linguistic Variable



What is low, medium, high? → We answer this with the membership function



↓
Some doctors call 100°F as low fever, some call it moderate.



Hence, membership function converts crisp input to fuzzy linguistic variables

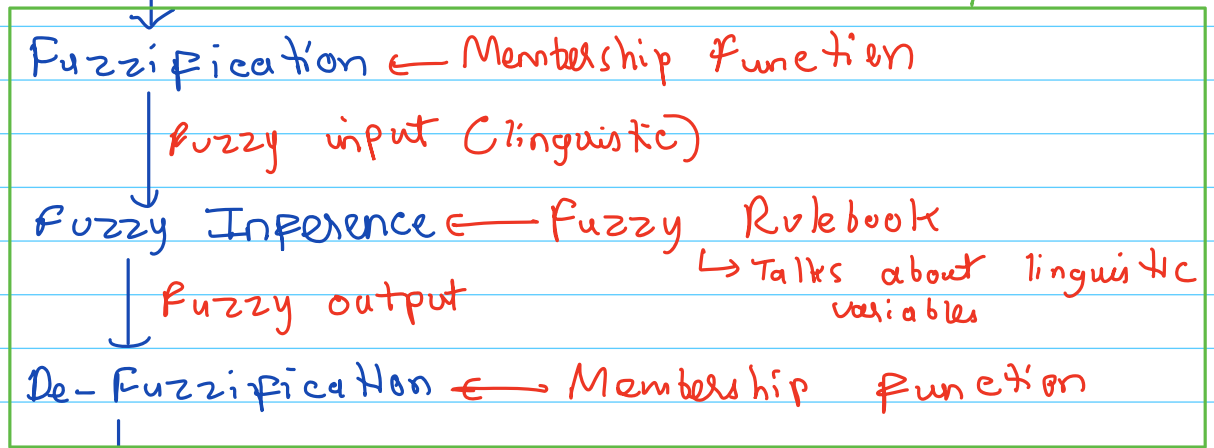
→ FUZZIFICATION

→ Reverse is called Defuzzification

Gross Algorithm

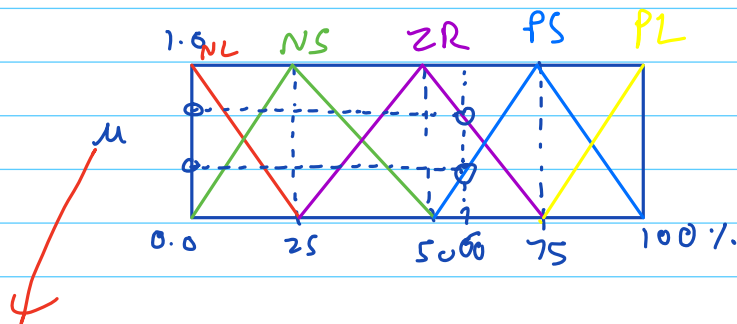
Crisp Numbers (I/P)

POZZY
PROCESSING



Crisp Numbers (O/P)

6/2/24



Crisp Input $\rightarrow$ 60%

Membership (we define it. There are no rules).

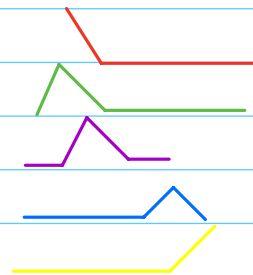
NL: Negative Large (v. low)

NS: Negative Small (low)

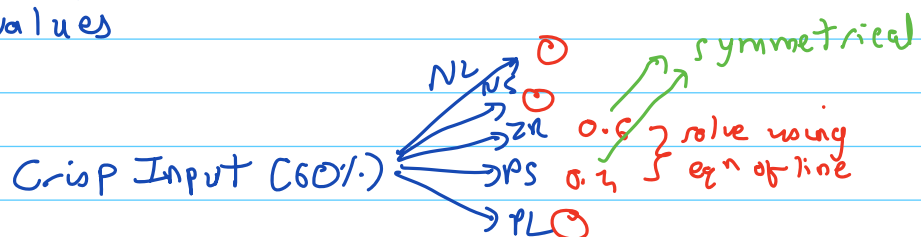
ZR: Zero (medium)

PS: Positive Small (high)

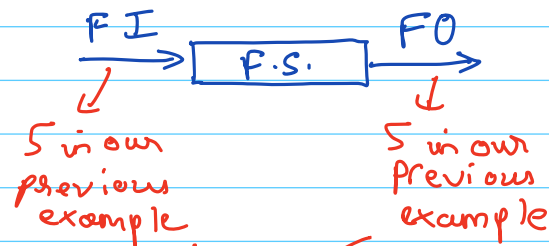
PL: Positive Large (v. high)



60% crisp input belongs to PS and ZR. Hence, the membership values



- For tuning purposes, we change the membership function curves.
- SISO: Single input, single output



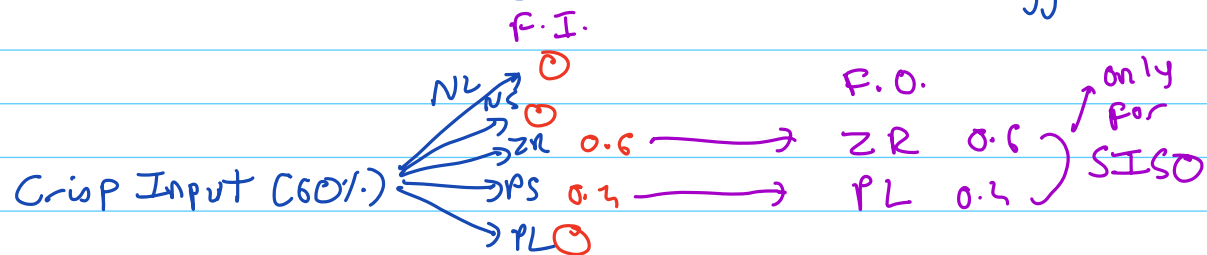
25 theoretical output combinations

Out of the 25, our combinations are decided by:-

RULE BOOK

Rule	IF (F.I)	THEN (F.O)
1	N.L.	N.L.
2	N.S.	Z.R.
3	Z.R.	Z.R.
4	P.S.	P.L.
5	P.L.	P.L.

From our 50% CI, Rule 3 and Rule 4 is triggered.



Note: Rule book is triggered only for non zero membership?.

Note: For 2 ISO

IF (X₁(μ₁) AND X₂(μ₂)) THEN (Y(μ))

NL (0.6)

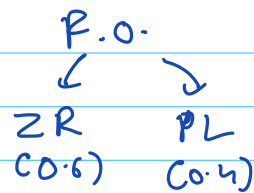
NS (0.2)

NL (0.2)

min(0.6, 0.2) → 0.2

"A chain is only as strong as its weakest link"

— Back to our example.



Defuzzification aggregates F.O. into C.O.

①

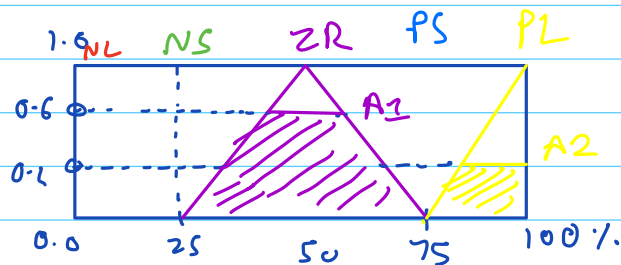
ZR (0.6) $\xrightarrow{\text{Highest Point (x co-ordinate)}}$ 50
 PL (0.4) $\xrightarrow{\hspace{1.5cm}}$ 100

} METHOD
OF
HEIGHT

Now, take weighted average.

$$50 \times 0.6 + 100 \times 0.4 = \boxed{70}$$

②



METHOD
OF
CG

$$A1 = \frac{1}{2} \times 50 \times 1 - \frac{1}{2} \times 0.4 \times 20 = 21$$

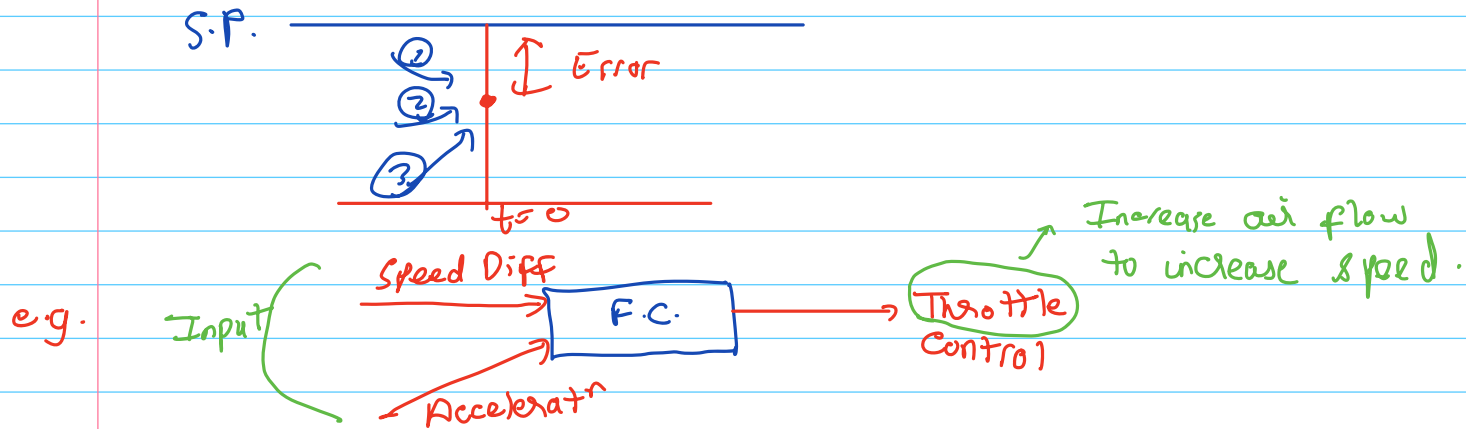
$$A2 = \frac{1}{2} \times 25 - \frac{1}{2} \times 0.6 \times 0.6 \times 25 = 9$$

$$CG \begin{cases} x_1 = 50 \\ x_2 = 100 - (25/3) = 91.66 \end{cases}$$

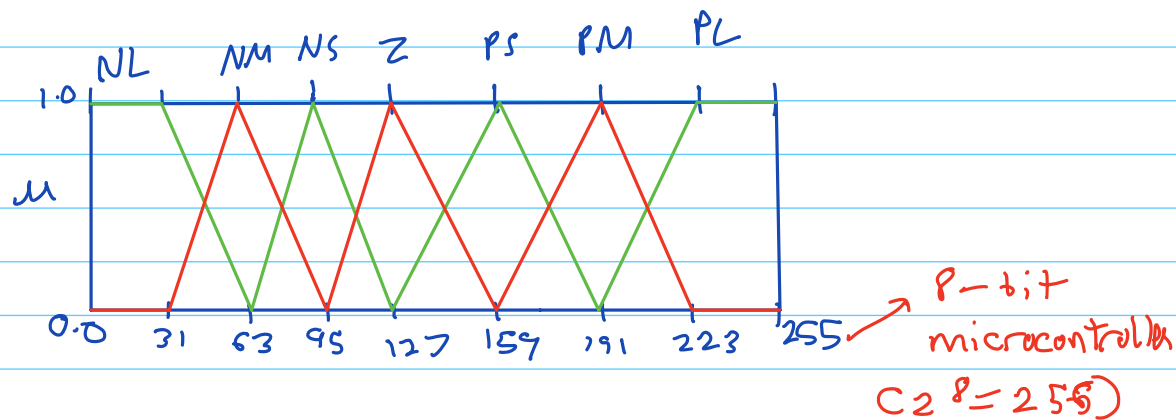
$$x^* = \frac{\sum A_i x_i}{\sum A_i} \rightarrow CO = 63.89$$

7/2/24

FUZZY CONTROLLER



Membership Function



N $\rightarrow$ -ve

P $\rightarrow$ +ve

L, M, S $\rightarrow$ Large, medium, small

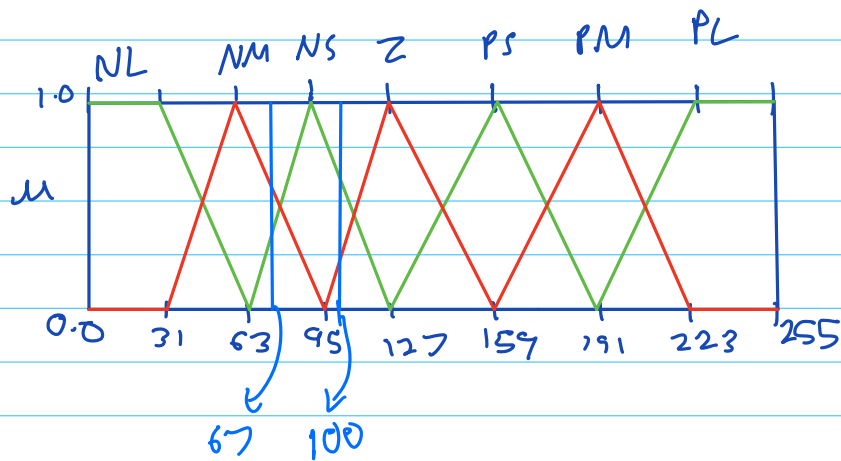
Let Speed Difference = 100
Acceleration = 67

Both normalised
b/w 0 to 255

- MISO

$\hookrightarrow$ Multi-input, single output.
C27

Step 1: Fuzzification



Speed Diff $\rightarrow$ ZR : 0.1563
 $\rightarrow$ NS : 0.8437

Acceleration $\rightarrow$ NS : 0.125
 $\rightarrow$ NM : 0.875

Step 2: Rule book

Rule #	IF (S.D.)	AND (CANCEL)	THEN (THROTTLE)
1	NL	Z	PL
2	Z	NL	PL
3	NM	Z	PM
4	NS	PS	PS
5	PS	NS	NS
6	PL	Z	NL
7	Z ①	NS	PS
8	Z ②	NM	PM

Out of 343

theoretical outputs

Input combinations

Z NS ① NS NM ③
 Z NM ② NS NM ④

Only Rule 7 and 8 will get triggered.

$$\begin{array}{l}
 Z-NS \Rightarrow PS \\
 \min(0.1563, 0.125) \rightarrow 0.125 \\
 Z-NM \Rightarrow PM \\
 \min(0.1563, 0.875) \rightarrow 0.1563
 \end{array}$$

minmax method $\rightarrow$ minimum input, maximum output
 $\downarrow$
 more than 1 output
 $\downarrow$
 eg C
 $PL \rightarrow 0.2$
 $PL \rightarrow 0.8$

Step 3: Defuzzification.

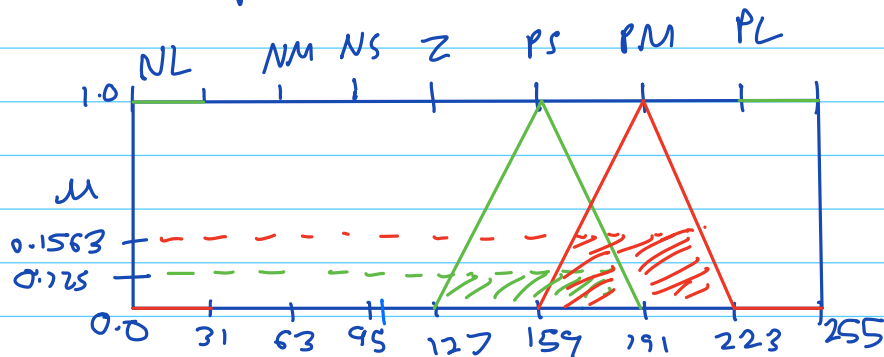
$$\begin{array}{l}
 PS \\
 (0.125) \\
 PM \\
 (0.1563)
 \end{array}$$

Method of heights

$$\begin{array}{l}
 PS: 159 \\
 PM: 191
 \end{array}$$

$$\Rightarrow CO = \frac{0.125 \times 159 + 0.1563 \times 191}{0.125 + 0.1563} = 176.7803$$

Method of CG



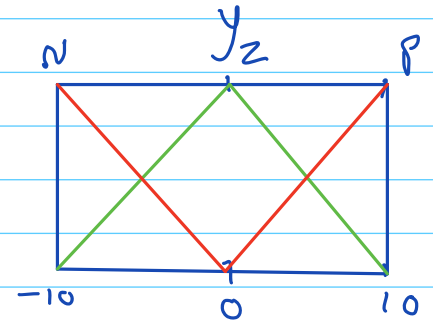
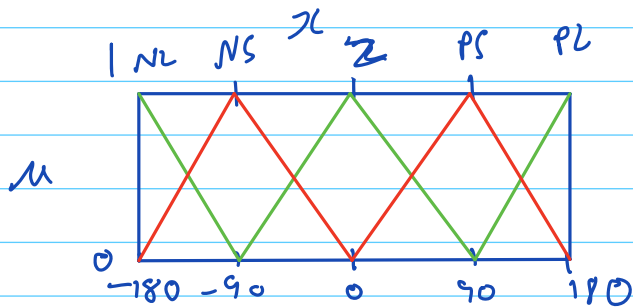
$$\frac{0.125 \times 64}{1}$$

$$APS = \frac{1}{2} \times 64 \times 1 - \frac{1}{2} \times 48 \times 0.875 = 11$$

$$APM = 0.5 \times 64 \times 1 - 0.5 \times 44 \times 0.84375 = 13.4375$$

$$\frac{\sum x_i A_i}{\sum A_i} = \frac{11 \times 159 + 13.4375 \times 191}{11 + 13.4375} = 176.5959$$

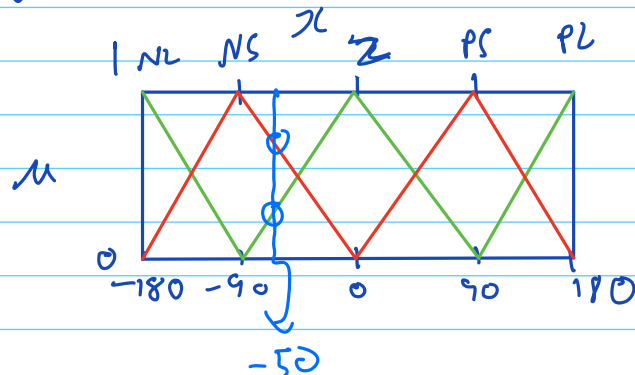
Q) $x \rightarrow \boxed{\text{SISO}} \rightarrow y$ $y = 10 \sin x$



Rule Book

x	NL	NS	z	PS	PL
y	z	N	z	P	z

$x = -50$



$x = -50 \rightarrow \begin{cases} z = 0.444 \\ NS = 0.556 \end{cases}$

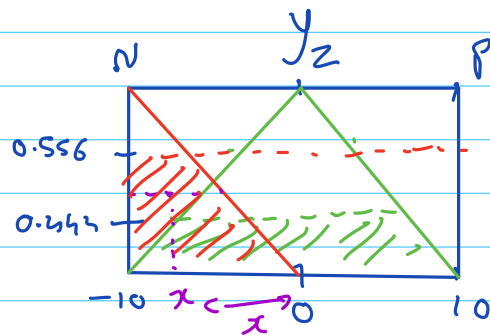
Rule book

$y \rightarrow \begin{cases} z = 0.444 \\ N = 0.556 \end{cases}$

Defuzzification

$$0.444 \times 0 + 0.556 \times (-10) / 1 = -5.56$$

CG



$$\frac{1}{2} \times (10 - x) \times \left(\frac{x}{10} + 1\right)$$

$$= \frac{1}{2} \times x \times \frac{x}{10}$$

$$A_z = \frac{1}{2} \times 20 \times 1 - \frac{1}{2} \times 2.24 \times 0.556$$

$$= 9.37728$$

$$x + 10 - \frac{x^2}{10} = \frac{x^2}{10}$$

$$\frac{x^2}{5} = 10$$

$$x = \sqrt{50}$$

$$= 7.071$$

$A_N =$

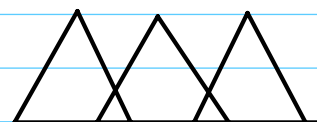
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Fuzzy logic

→ Rulebook (Intelligence)

→ Membership function (tuning parameter, can be changed to a different forms)

(*) if M.F is non-uniform
the sum may not be 1.



main tasks
during Fuzzification
Algorithm.

Mathematical model

1.) PDE / Equations

2.) B.C / I.C.

3.) K, ρ, C_p (may vary with phase change)

4.) Assumptions

↳ continuous casting

Take a lot of time (not employed in Real Time systems)

∴ we use AI systems (takes time to train, but once trained, decision processing takes less time).

Gradient-free optimisation (AI types)

↳ GA (Genetic Algorithm).

↳ PSO

↳ ACO (Ant colony optimisation)

↳ SA (simulated annealing)

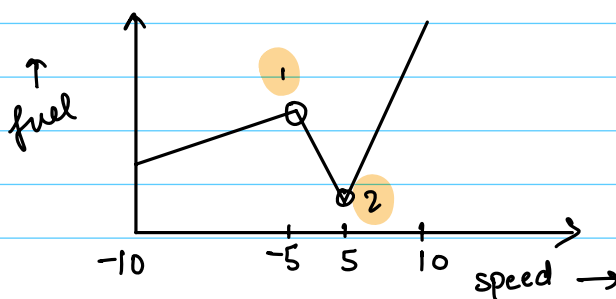
Fuzzy Rule set

Non-fuzzy rule set example:-

if $-10 \leq x \leq -5$ then $y = 0.1x + 7.0$ — ①

if $-5 \leq x \leq +5$ then $y = -0.5x + 4$ — ②

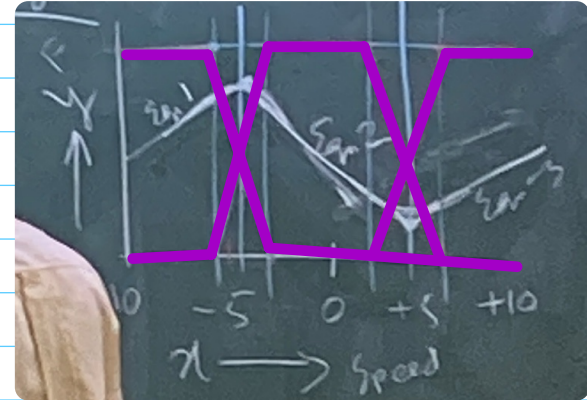
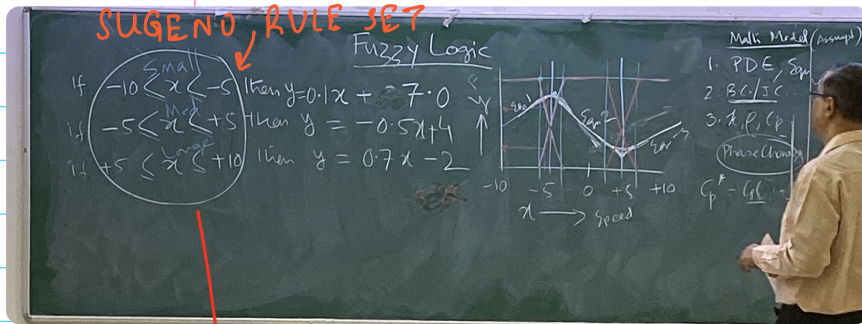
if $+5 \leq x \leq 10$ then $y = 0.7x - 2$ — ③



[Multi-point fuel injection] systems
(These curves are generated from OFFLINE simulations)

of Thermodynamic, Torque generation etc. data)

At point 1 & 2 the car engine gets jerk due to sudden change in fuel flow rate. (SMOOTHING is needed)



Fuzzy rule set
(smoothing of sharp corner)

membership function values will be weights for eq'n ①, ②, ③

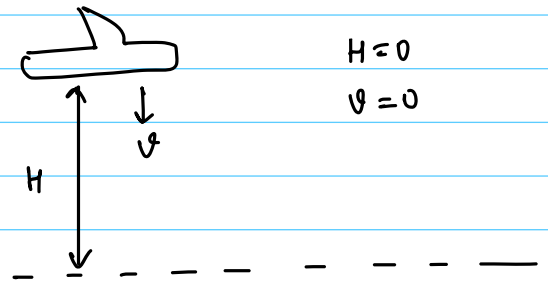
If $\sum u_i = 1$ then result is $\sum u_i x_i$

$\sum u_i \neq 1$ then $\frac{\sum u_i x_i}{\sum u_i}$

Our original rule set method (fuzzy $\rightarrow$ fuzzy)
if x is (small) then y is (small) .

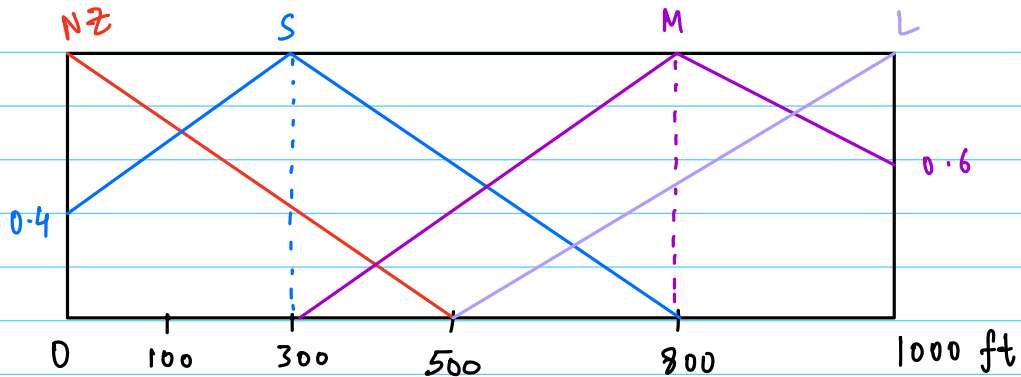
"Mamdani"

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$v \rightarrow$ FLC $\rightarrow$ control force (to control speed in y-direction)
 $H \rightarrow$

M.F
(Height)



Nz - near zero

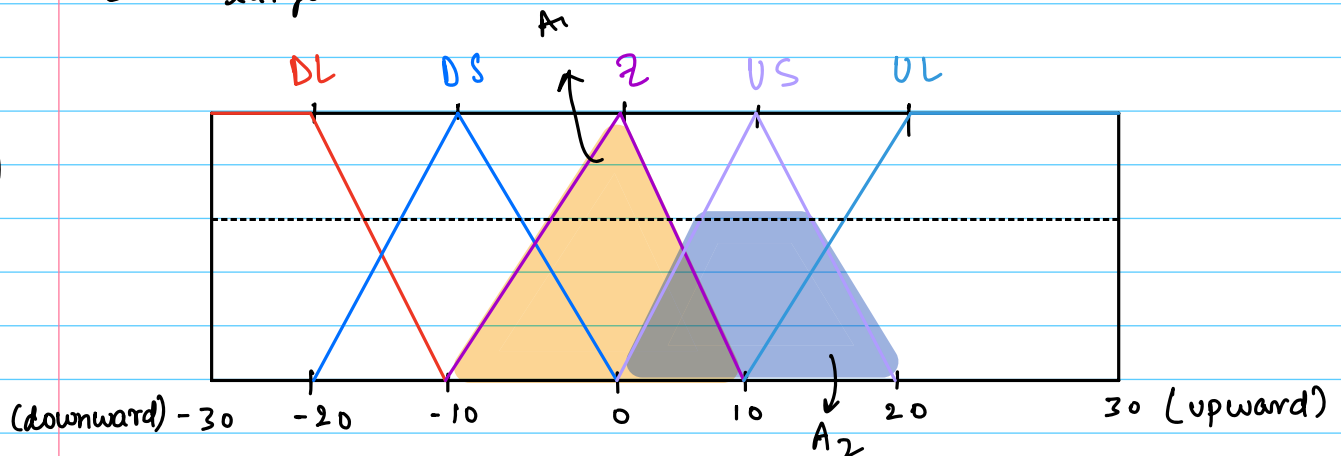
S - small

M - medium

L - large

$[\sum u_i \neq 1 \text{ (asymmetrical m.f)}]$
(may)

M.F
(v, F)



$v \rightarrow$ ft/s

$F \rightarrow$ force limits

Rule Base

$\begin{matrix} v, f \\ h \end{matrix}$	DL	DS	Z	US	UL
L	Z	DS	DL	DL	DL
M	US	Z	DS	DL	DL
S	UL	US	Z	DS	DL
Z	UL	UL	Z	DS	DS

If (H=M) AND (V=Z) THEN (F=DS)

$$F = ma$$

$$f = \frac{F}{m} = \frac{\Delta v}{\Delta t} \Rightarrow \boxed{f = \Delta v}$$

$$\text{If } \Delta t = 1 \text{ sec}$$

$$\Rightarrow v_{i+1} = v_i + f_i$$

$$\Rightarrow h_{i+1} = h_i + v_i$$

$$\text{Ex: } h_0 = 1000 \text{ ft}$$

$$v_0 = -20 \text{ ft/s}$$

$$F_0 = ? \text{ (from FLC)}$$

$$\begin{aligned} \text{(\#)} \quad h_0 &= 1000 \text{ ft} \begin{cases} \rightarrow L(1) \\ \rightarrow M(0.6) \end{cases} \\ v_0 &= -20 \text{ ft/s} \rightarrow DL(1) \end{aligned}$$

$$\Rightarrow F \begin{cases} \rightarrow Z(1) \\ \rightarrow US(0.6) \end{cases}$$

2 rules triggered as highlighted

$$A_1 = \frac{1}{2} \times 1 \times 20 = 10$$

$$A_2 = \frac{1}{2} \times 1 \times 20 - \frac{1}{2} \times 0.4 \times 8 = 8.4$$

$$\text{Crisp Output} = \frac{\sum A_i x_i}{\sum A_i} = \frac{10(0) + 8.4(10)}{10 + 8.4}$$

$$= \underline{\underline{4.5652 (x^*)}}$$

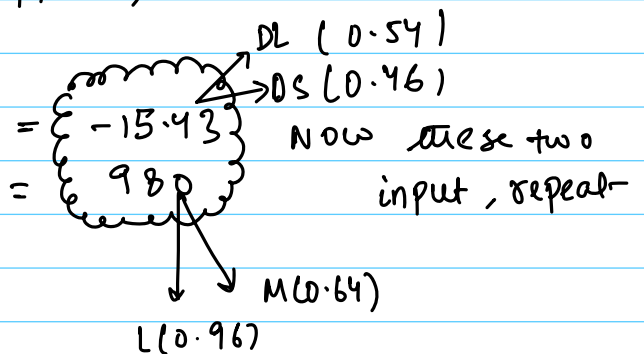
Method of height

$$(C.O) = \frac{(1 \times 0) + (0.6 \times 10)}{(1 + 0.6)} = 3.75 \text{ (2*)}$$

(#)

$$v_1 = -20 + 4.57 = -15.43$$

$$h_1 = 1000 - 20 = 980$$



Input

Output

min	L, DL	→	Z (0.54)	} →	DS 0.46 US 0.54 Z 0.54 max
min	L, DS	→	DS (0.46)		
min	M, DL	→	US (0.54)		
min	M, DS	→	Z (0.46)		

$$M/H \rightarrow 0.52$$

$$M/LG \rightarrow 0.35$$

$$v_2 = -15.43 + 0.35 = -15.08$$

$$h_2 = 980 - 15.43 = 964.57$$